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26 September 2025

The Basics of Neural Networks

Neural networks are artificial intelligence machines that are modeled after a human brain. Most neural networks have three basic parts: an input layer, at least one middle layer (although there's usually more), and an output layer. The input layer is the first layer to receive data and process it; this data can be a wide variety of things from images to large amounts of texts. Then the input layer sends the data to the many hidden middle layers, where mathematical operations are done to determine the strength of the connection between the many sets of neurons, called a weighted connection. As the data goes through the layers, the network becomes more fine-tuned in a process called training, creating a model that is accurate and can identify patterns and relationships within the data. After the data has gone through all of the middle layers, it will go to the output layer, where the final output is produced. This entire process can be done multiple times with different parameters to improve the system's ability to process and classify data.

Using Neural Networks for Personalization

There have been many applications for neural networks over the years, but none have as many ethical implications as personalizing user experiences. When personizing a user's experience, neural networks will process large amounts of data about that user's behavior (i.e. user purchases, likes and dislikes on posts) and then create a predictive model that can make

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recommendations or offer content that is tailored to the user's preferences. Due to this process of collecting personal information, these algorithms raise concerns for consumer privacy; people often don't fully understand how their personal data is being used and may not have as much control over data they may or may not want others to have access to. Additionally, because users cannot see the algorithm itself, many types of bias can easily slip under the radar, making it difficult for users to pinpoint unfair outcomes.

Personalization under GDPR Laws

The General Data Protection Regulation (GDPR) are European Union (EU) laws that regulate what companies can and cannot do with a user's personal data, which allows for greater privacy and provide users with more control over their personal information. The GDPR affects personalization in a number of ways:

- Transparency: The GDPR requires companies to provide consumers with clear information about how their data will be used for personalization,
- Purpose limitation: The companies cannot use personal data for any purpose other than those that have been disclosed to users,
- Data minimization: The GDPR also requires that companies can only collect the minimum amount of data necessary to complete the personalization objectives, and

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- Storage limitation: Companies cannot retain personal data for longer than is necessary to achieve their personalization objectives.

Company Practices under GDPR Laws

The social networking company may be violating the GDPR through a lack of transparency as well as data minimization. This is due to the fact that the company's neural networks are obtaining and storing information on everything the user does on the app, from what they click on to how much time is spent on the app. Some of this data could be more than what the company needs for their app to function. Additionally, while users may be aware that their personal information is being collected, they may not know what kind of data is being collected or have a clear pathway to have that information removed if they don't want the company to store it anymore. Because the company wants their app to work by "remembering" user preferences and settings, they still have to collect some data. That being said, users should have the right to control how their personal information is being used and stored, and the company needs to respect those rights in their practices.

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networks#:~:text=On%20a%20high%20level%2C%20the,model%20with%20a%20constrained%20output.